# LAB NOC 07

## Cisco 888 ISR — Baseline Reset / Credential Recovery

Real Hardware | ES / EN | NOC Portfolio

### INFORMACION DEL LABORATORIO / LAB INFORMATION

| | | |
|---|---|---|
| **Hardware / Hardware** | Cisco 888 Integrated Services Router (ISR) | Cisco 888 Integrated Services Router (ISR) |
| **Ticket ID** | NOC-LAB-07 | NOC-LAB-07 |
| **Categoria / Category** | Acceso administrativo — recuperacion de credenciales | Administrative access — credential recovery |
| **Nivel / Level** | CCNA — Router management / IOS fundamentals | CCNA — Router management / IOS fundamentals |
| **Duracion / Duration** | Aprox. 30 minutos | Approx. 30 minutes |
| **Acceso / Access** | Consola serial — PuTTY COM5, 9600 bps, 8N1 | Serial console — PuTTY COM5, 9600 bps, 8N1 |
| **Entorno / Environment** | Hardware real (equipo usado con config previa) | Real hardware (used equipment with prior config) |
| **Relevancia NOC** | Integracion de equipos usados, factory reset, baseline | Used equipment integration, factory reset, baseline |

### OBJETIVO / OBJECTIVE

| OBJETIVO | OBJECTIVE |
|---|---|
| Recuperar el acceso administrativo a un router Cisco 888 con credenciales desconocidas, ejecutar un factory reset completo via consola serial, y verificar que el equipo quedo en estado limpio (baseline) listo para configuracion inicial — documentando el proceso en formato de ticket NOC. | Recover administrative access to a Cisco 888 router with unknown credentials, perform a complete factory reset via serial console, and verify that the equipment is in a clean state (baseline) ready for initial configuration — documenting the process in NOC ticket format. |

### INCIDENT TICKET — NOC-LAB-07

| CAMPO / FIELD | DETALLE / DETAIL |
|---|---|
| **Ticket ID** | NOC-LAB-07 |
| **Type / Tipo** | Router con credenciales desconocidas — acceso administrativo bloqueado / Router with unknown credentials — administrative access blocked |
| **Priority / Prioridad** | High — equipo sin acceso no puede integrarse a produccion / High — inaccessible equipment cannot be integrated into production |
| **Device / Equipo** | Cisco 888 Integrated Services Router (ISR) |

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| **Access Method** | Consola serial — PuTTY, COM5, 9600 bps, 8N1, Flow Control: None |
| **Symptom / Sintoma** | Router solicita usuario y contrasena al iniciar sesion — credenciales previas desconocidas / Router prompts for username and password — previous credentials unknown |
| **Diagnosis** | Startup-config almacenada en NVRAM con autenticacion configurada / Startup-config stored in NVRAM with authentication configured |
| **Interrupt Method** | Reinicio + Special Command > Break (clic derecho en PuTTY durante el boot) / Reboot + Special Command > Break (right-click in PuTTY during boot) |
| **Resolution** | enable → erase startup-config → reload → confirmar sin guardar / enable → erase startup-config → reload → confirm without saving |
| **Verification** | show running-config (sin IPs, sin usuarios) + show startup-config (startup-config is not present) |
| **Final State** | Router# prompt limpio — equipo listo para configuracion inicial / Clean Router# prompt — equipment ready for initial configuration |
| **Status** | CERRADO — baseline completado / CLOSED — baseline completed |

## PROCEDIMIENTO — 4 ETAPAS / PROCEDURE — 4 STAGES

| # | ESPANOL | ENGLISH |
|---|---|---|
| 1 | ETAPA 1 — Recibir ticket: router Cisco 888 con credenciales desconocidas — acceso bloqueado | STAGE 1 — Receive ticket: Cisco 888 router with unknown credentials — access blocked |
| 2 | Conectar cable de consola USB-RJ45 al puerto de consola del Cisco 888 | Connect USB-RJ45 console cable to the Cisco 888 console port |
| 3 | Verificar puerto COM en Administrador de dispositivos → COM5 | Verify COM port in Device Manager → COM5 |
| 4 | Abrir PuTTY: Serial, COM5, 9600 bps, 8N1, Flow Control: None | Open PuTTY: Serial, COM5, 9600 bps, 8N1, Flow Control: None |
| 5 | ETAPA 2 — Interrumpir el boot: desenchufar y volver a enchufar el router | STAGE 2 — Interrupt boot: unplug and replug the router |
| 6 | Durante el arranque: clic derecho en PuTTY → Special Command → Break | During boot: right-click in PuTTY → Special Command → Break |
| 7 | Router cae a prompt privilegiado — omitiendo autenticacion de startup-config | Router drops to privileged prompt — bypassing startup-config authentication |
| 8 | ETAPA 3 — Factory reset: ejecutar enable → erase startup-config → reload | STAGE 3 — Factory reset: run enable → erase startup-config → reload |
| 9 | Confirmar NO guardar la configuracion cuando el sistema lo solicite durante el reload | Confirm NOT to save configuration when system prompts during reload |
| 10 | Observar en syslog: SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram | Observe in syslog: SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram |
| 11 | ETAPA 4 — Verificacion: show running-config → sin IPs, sin usuarios, config minima | STAGE 4 — Verification: show running-config → no IPs, no users, minimal config |

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|---|---|---|
| 12 | Verificar con show startup-config → respuesta: 'startup-config is not present' — evidencia definitiva | Verify with show startup-config → response: 'startup-config is not present' — definitive evidence |
| 13 | Confirmar prompt: Router# — equipo en estado de fabrica, listo para baseline | Confirm prompt: Router# — equipment in factory state, ready for baseline |
| 14 | Cerrar ticket NOC-LAB-07 con evidencia documentada (outputs de show) | Close ticket NOC-LAB-07 with documented evidence (show outputs) |

## METODO DE INTERRUPCION DE BOOT / BOOT INTERRUPT METHOD

| # | ACCION (ES) | ACTION (EN) |
|---|---|---|
| 1 | Conectar cable de consola USB-RJ45 al router | Connect USB-RJ45 console cable to router |
| 2 | Abrir PuTTY — COM5, 9600 bps, 8N1, Flow Control: None | Open PuTTY — COM5, 9600 bps, 8N1, Flow Control: None |
| 3 | Desenchufar y volver a enchufar el router (cycle power) | Unplug and replug the router (cycle power) |
| 4 | Durante el arranque: clic derecho en la barra de PuTTY → Special Command → Break | During boot: right-click on PuTTY title bar → Special Command → Break |
| 5 | El router interrumpe el boot y cae a modo ROMMON o prompt privilegiado | Router interrupts boot and drops to ROMMON or privileged prompt |
| 6 | Ejecutar: enable (si requiere modo privilegiado) | Run: enable (if privileged mode required) |

## RESOLUCION PASO A PASO / STEP-BY-STEP RESOLUTION

| # | COMANDO / COMMAND | ACCION (ES) | ACTION (EN) |
|---|---|---|---|
| 1 | `enable` | Ingresar a modo EXEC privilegiado | Enter privileged EXEC mode |
| 2 | `show running-config` | Verificar config actual antes de borrar (documentar) | Verify current config before erasing (document) |
| 3 | `erase startup-config` | Borrar la NVRAM — eliminar toda configuracion previa | Erase NVRAM — remove all previous configuration |
| 4 | `reload` | Reiniciar el router para aplicar el borrado | Reload the router to apply the erase |
| 5 | `[Confirmar: no guardar]` | Cuando pregunte si guardar: responder NO | When asked to save: answer NO |
| 6 | `show running-config` | Verificar: sin IPs, sin usuarios, sin configuracion previa | Verify: no IPs, no users, no previous configuration |
| 7 | `show startup-config` | Verificar: 'startup-config is not present' — evidencia definitiva | Verify: 'startup-config is not present' — definitive evidence |

## REFERENCIA DE COMANDOS / COMMAND REFERENCE

| COMANDO / COMMAND | SIGNIFICADO (ES) | MEANING (EN) |
|---|---|---|

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|---|---|---|
| `enable` | Ingresar a modo EXEC privilegiado (Router> → Router#) | Enter privileged EXEC mode (Router> → Router#) |
| `show running-config` | Muestra la configuracion activa en RAM — verifica estado actual | Shows active configuration in RAM — verifies current state |
| `show startup-config` | Muestra la config guardada en NVRAM — clave para confirmar factory reset | Shows config saved in NVRAM — key to confirm factory reset |
| `erase startup-config` | Borra la NVRAM completamente — elimina todas las credenciales y configuracion | Completely erases NVRAM — removes all credentials and configuration |
| `reload` | Reinicia el router — aplica el borrado de NVRAM | Restarts the router — applies NVRAM erase |
| `show version` | Muestra modelo, IOS version, NVRAM y memoria — util para inventario | Shows model, IOS version, NVRAM and memory — useful for inventory |

## EVIDENCIA DE VERIFICACION / VERIFICATION EVIDENCE

| MENSAJE / COMANDO | OUTPUT OBSERVADO | INTERPRETACION / INTERPRETATION |
|---|---|---|
| `SYS-7-NV_BLOCK_INIT` | `Initialized the geometry of nvram` | NVRAM reinicializada correctamente durante el reload / NVRAM correctly reinitialized during reload |
| `show running-config` | `No IP addresses, no users, minimal system config only` | Config limpia — sin credenciales, sin interfaces configuradas / Clean config — no credentials, no configured interfaces |
| `show startup-config` | `startup-config is not present` | Evidencia definitiva de NVRAM limpia — equipo en estado de fabrica / Definitive evidence of clean NVRAM — factory state |
| `Router# prompt` | `Router# (sin nombre personalizado)` | Prompt limpio confirma ausencia de hostname configurado / Clean prompt confirms no configured hostname |

## TROUBLESHOOTING DOCUMENTADO / DOCUMENTED TROUBLESHOOTING

| HALLAZGO / FINDING | ACCION CORRECTIVA / CORRECTIVE ACTION |
|---|---|
| El router solicitaba usuario y contrasena al acceder por consola — credenciales de configuracion previa desconocidas. / Router prompted for username and password on console access — unknown previous configuration credentials. | Se utilizo el metodo de interrupcion de boot via Special Command > Break en PuTTY durante el encendido. Este metodo es la tecnica estandar para recuperacion de acceso en Cisco IOS sin conocer las credenciales. / Boot interruption method used via Special Command > Break in PuTTY during power-on. This is the standard Cisco IOS access recovery technique without knowing credentials. |

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| Despues del erase startup-config el sistema pregunta si se desea guardar la configuracion antes del reload. / After erase startup-config the system asks whether to save configuration before reload. | Responder NO es critico — si se responde YES se vuelve a guardar la configuracion que se queria borrar, revirtiendo todo el proceso. / Answering NO is critical — answering YES saves back the configuration that was being erased, reverting the entire process. |
| El tip profesional: muchos ingenieros solo verifican con show running-config pero no ejecutan show startup-config. / Professional tip: many engineers only verify with show running-config but do not run show startup-config. | El output 'startup-config is not present' es la evidencia definitiva de que la NVRAM esta limpia. Sin este paso el ticket no tiene prueba completa del factory reset. / The output 'startup-config is not present' is the definitive evidence that NVRAM is clean. Without this step the ticket lacks complete proof of the factory reset. |

## RESULTADO / RESULT

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| **RESULTADO (ES)**<br>Acceso recuperado mediante interrupcion de boot. Factory reset ejecutado con erase startup-config. NVRAM limpia verificada con show startup-config ('not present'). Router# prompt limpio. Equipo listo para configuracion inicial. Ticket NOC-LAB-07 cerrado. | **RESULT (EN)**<br>Access recovered via boot interruption. Factory reset executed with erase startup-config. Clean NVRAM verified with show startup-config ('not present'). Clean Router# prompt. Equipment ready for initial configuration. Ticket NOC-LAB-07 closed. |

### KEY TAKEAWAY

*"El factory reset de un router es una habilidad de emergencia critica en NOC. Los dos comandos de verificacion no son opcionales: show running-config muestra que no hay configuracion activa, pero show startup-config con respuesta 'not present' es la prueba forense definitiva de que la NVRAM esta limpia. Sin ese segundo comando el ticket no esta completo."*

*"Router factory reset is a critical emergency skill in NOC. The two verification commands are not optional: show running-config shows there is no active configuration, but show startup-config with 'not present' response is the definitive forensic proof that NVRAM is clean. Without that second command the ticket is incomplete."*

### NOC RELEVANCE — MINING / INDUSTRIAL ENVIRONMENT

In a mining operation (e.g., Newmont Cerro Negro), routers managing WAN links, VPN tunnels, or remote site connectivity may become inaccessible due to unknown credentials — whether from staff turnover, undocumented changes, or acquired second-hand equipment. A NOC engineer must be able to: (1) perform console-based access recovery without relying on remote access (SSH/Telnet may be unavailable), (2) execute a controlled factory reset without disrupting other equipment on the same console server, (3) document the full procedure for compliance and audit purposes, and (4) hand off a clean-baseline device to the configuration team. This lab directly represents the first step of router deployment in a NOC environment.